Time Series Analysis of Spotify Song Features

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Spotify®

Project Details

Motivation

- Certain songs become prevalent in certain periods of time… we wanted to understand WHY this happens. Are there any trends we could find based on around when in the year the songs released?
    - Ex: Slower songs in Winter months and more upbeat songs in the Summer months.

Hypothesis

- There exists “seasonality”, or some kind of significant yearly pattern with at least one feature and not all features of songs based on their month of release.

Research Question

- Are there seasonal patterns in song features (energy, danceability, valence, etc.) based on month of release?

Modeling Approach

- Spotify Data → Preprocess → SARIMA → Stationarity (if needed) → Best-Fitting Model → Evaluate

GOAL

- Use Spotify song data to see if certain song features tend to appear “more” or “less” based on the month of the year.
    - If a feature has a consistent pattern, then we will determine that it has seasonal behavior.

Data Explanation and Acquisition

- **Acquisition**: We obtained our dataset on Github from an existing Spotify data analysis project.

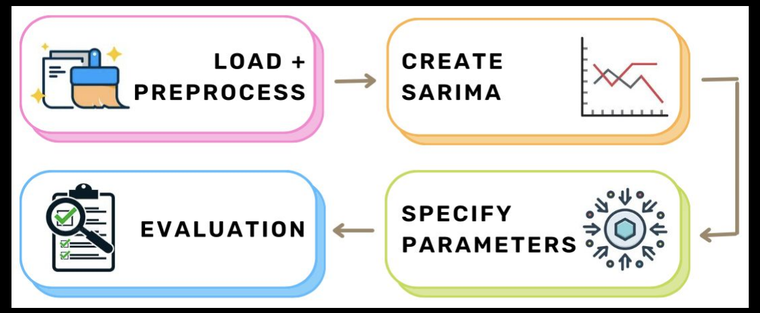
- **Size**:
  - Originally: 169909 rows × 19 columns
  - Cleaned: 68369 rows × 8 columns
    - Reduced to only last 40 years
    - Dropped columns not considered
    - Removed rows without months
    - New month column
  - New Dataset: 486 rows x 5 columns
    - Avg. feature per month each year

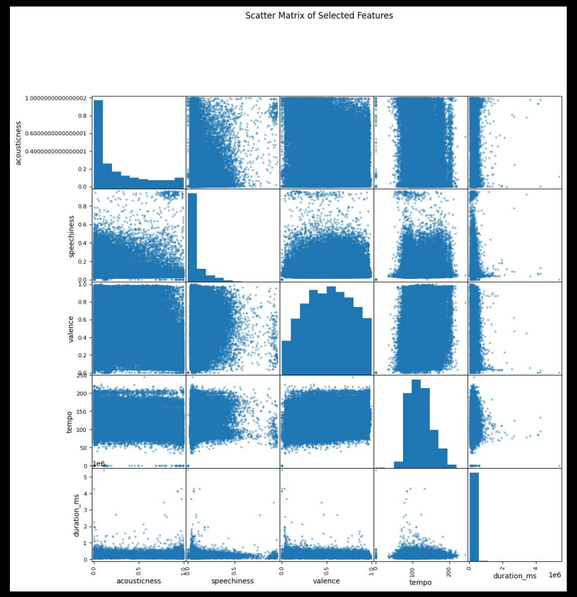
year	month	avg_acousticness	avg_duration	avg_speechiness	avg_valence	avg_tempo
1980	1	0.43426417289156627	268339.9397590361	0.05929156626506024	0.5653346385542168	119.47609638554216
1980	2	0.20926071459459458	229122.98648648648	0.05223108108108108	0.648445945945946	128.68493243243242
1980	3	0.18687820759493673	243819.8987341772	0.05816708860759494	0.5132683544303798	127.4242658227848
1980	4	0.26999417297297296	236968.14864864864	0.06580270270270269	0.6096648648648649	122.25082432432433
1980	5	0.2867609782608696	258279.70652173914	0.04970652173913044	0.7149565217391304	125.67446739130435
1980	6	0.24773316666666667	239664.72727272726	0.06298636363636363	0.5876333333333333	118.0205
1980	7	0.18393497916666668	275837.8541666667	0.05975833333333334	0.6209791666666666	128.75570833333333
1980	8	0.2955457741935484	266341.0	0.05938709677419355	0.5379354838709677	126.96458064516129
1980	9	0.333910712	248746.024	0.06494159999999999	0.5820080000000001	117.114168
1980	10	0.21647928385093168	269484.9254658385	0.06326583850931677	0.6571869565217392	125.71005590062111
1980	11	0.22194902362204724	232230.3622047244	0.05963385826771653 4	0.504371653543307	120.72315748031497
1980	12	0.34451735263157895	261384.70175438595	0.06953859649122807	0.5720333333333333	121.04228070175438
1981	1	0.3546523733799534	259120.3682983683	0.06387412587412587	0.5829631701631701	121.21100932400932
1981	2	0.3401822	279877.775	0.0512475	0.46607750000000003	110.67204999999998
1981	3	0.4481189189189189	305574.54054054053	0.05842432432432432	0.6324189189189189	112.46243243243242
1981	4	0.3538138276315789	282460.7236842105	0.050410526315789475	0.5656434210526315	125.20739473684212
1981	5	0.20517980793650795	299181.44444444444	0.05819047619047619	0.5887698412698413	119.47893650793651
1981	6	0.07843448979591836	246844.5306122449	0.05446734693877551	0.6112183673469388	138.3072857142857
1981	7	0.1531651024691358	259387.03703703705	0.0618	0.5748333333333333	[illegible]9506

IntelliJ IDEA CE

Analysis Plan & Justification

- Perform EDA to identify patterns and spread of data
- Preprocess Data
  - 169909 row × 19 col → 486 row x 5 col
  - Dataset of avg. feature per month each year
- Create SARIMA model
  - One model per song feature to look at seasonal patterns individually
  - Season period of 12 months (1 year)
- Specify appropriate model parameters
  - Identify stationary vs nonstationary; transform to stationary if not
  - Test seasonal and nonseasonal params to find best model per feature (lowest AIC/BIC)
- Evaluate results
  - Create visualizations to help see patterns
  - Statistically significant seasonal parameters

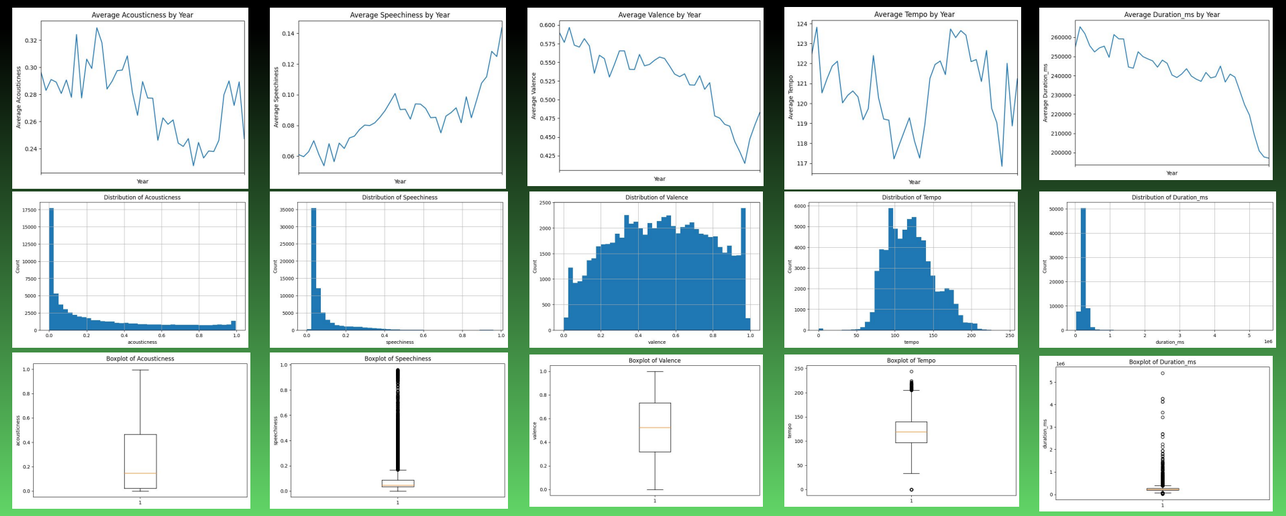




Avg. val per year

Distribution

Boxplot



Tricky Analysis Decision

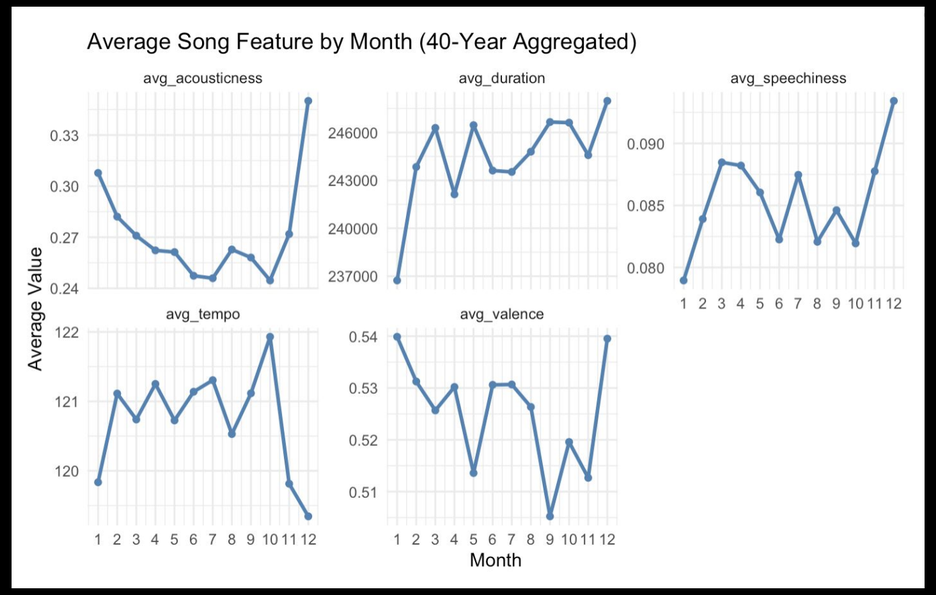
- When attempting to find the best parameter combination for the best-fitting model, running the code for all combinations was extremely time-consuming
- So for general effectiveness, we chose to use stepwise + approximation to reduce time while also still finding relatively strong performing models
- However, we have to acknowledge that there is a chance we might have missed the true, top-performing model, so during our second evaluation process (with Python) we went through all parameter combinations

Bias and Uncertainty Validation

- May be some selection bias from original dataset:
  - Though it contains several decades of song data, there's not a definite way to confirm that data represents all music genres across the world
- Additionally, we might want to keep in mind that music listening habits vary according to trends and music production, so that might factor in to any results that we produce.
- Again, the first round of evaluation in R did cover all parameter combinations, so there is uncertainty there involving the best-fitting model
  - Addressed during second evaluation

Results

Feature	Best Order	Best Seasonal Order	AIC	BIC	p-value	Season Structure	Significant Seasonal Terms
Acousticness	(2, 1, 1)	(1, 0, 1, 12)	-1187.6472359536715	-1162.718087389573	0.00188997959988896 54	True	2
Speechiness	(1, 0, 1)	(0, 0, 0, 12)	-2246.629313234098	-2234.083058513948	0.8197318483395217	False	0
Valence	(1, 0, 1)	(0, 0, 0, 12)	-1532.3893369717812	-1519.8430822516314	0.5798530109658852	False	0
Tempo	(1, 1, 2)	(0, 1, 1, 12)	2649.885266512354	2670.519612432925	1.6855546053936836e-13	True	1
Duration	(2, 0, 2)	(1, 1, 1, 12)	10458.341763669152	10487.245115139576	0.9855625229592424	True	2



Conclusion

Looking at our final evaluation in Python, there **IS** substantial evidence of seasonality with **acousticness**, **tempo**, and **duration** with at least one significant parameter!

There is **NOT** enough evidence of **speechniess** or **valence** being seasonal.

Next Steps

1. Back to R

Like we did for the second round evaluation in Python, we might want to try all parameter combinations in R as well to double check our code and model results.

2. More Features

Expand explored features and year range during analysis (danceability, 1970s, etc).

3. Forecasting

Consider predictive forecasting with SARIMA, which will help us predict future trends for each song feature.

References

[1] calebelgut, "GitHub - calebelgut/spotify-lstm: Classification & Time Series Analysis of Spotify Data using Grid Searched Random Forest & LSTM.," *GitHub*, 2025. https://github.com/calebelgut/spotify-lstm (accessed Feb. 25, 2026).

[2] Mark Mulligan, "Music subscriber market shares Q2 2021," *MIDiA*, Jan 18, 2022. https://www.midiaresearch.com/blog/music-subscriber-market-shares-q2-2021#:~:text=Spotify%20remains%20the%20DSP%20with,for%20the%20second%20successive%20year. (accessed Feb. 27, 2026).

[3] Xiaolei Liu, Zi Lin, Ziming Feng, "Short-term offshore wind speed forecast by seasonal ARIMA - A comparison against GRU and LSTM," *Energy (Volume 227)*, Jul 15, 2021. https://www.sciencedirect.com/science/article/abs/pii/S0360544221007416#sec6 (accessed Feb. 27, 2026).

[4] Paliari et al., "A comparison of the optimized LSTM, XGBOOST and ARIMA in Time Series forecasting," *2021 12th International Conference on Information, Intelligence, Systems & Applications (IISA)*, 2021. https://ieeexplore.ieee.org/abstract/document/9555520 (accessed Feb. 27, 2026).

[5] Jason Brownlee, "A Gentle Introduction to SARIMA for Time Series Forecasting in Python," *Machine Learning Mastery*, Aug. 16, 2018. https://machinelearningmastery.com/sarima-for-time-series-forecasting-in-python/

[6] Kontopoulou et al., "A Review of ARIMA vs. Machine Learning Approaches for Time Series Forecasting in Data Driven Networks," *Future Internet*, 2023. https://www.mdpi.com/1999-5903/15/8/255 (accessed Feb. 27, 2026).

Group 1's | Favorite | Song/Album

Questions?

兰亭序 (Orchid Pavilion Preface)

黑马 (The Dark Horse)

All Things Must Pass

Alive 2007

Merry-Go-Round of Life

Departure (Samurai Champloo)